

MAINTENANCE WORK ORDER

Corrective Maintenance – Bearing Assembly Replacement

Work Order No.	WO-2024-1187	Date Issued	14-Nov-2024
Equipment	Centrifugal Pump P-204A	Location	Process Bay 2
Work Type	Corrective	Reference SOP	SOP-PM-204
Assigned Technician	Rajesh Kumar (Senior Maintenance Engineer)	Shift	Day Shift
Permit Required	General Work Permit (SOP-PTW-01)	Permit No.	PTW-2024-3312

Reason for Work Order

Control room recorded a sustained high-vibration alarm on Pump P-204A during routine monitoring on 13-Nov-2024. Shift supervisor raised a corrective work order the following day per standard practice for any vibration alarm exceeding the threshold specified in SOP-PM-204. Permit to Work PTW-2024-3312 was approved prior to isolation of the equipment.

Description of Work Performed

Pump P-204A was isolated per approved Permit to Work. On inspection, the bearing assembly showed visible wear consistent with the recorded vibration signature. The worn bearing assembly was removed and replaced in full. Mechanical seal was inspected during the same opening and found to be in acceptable condition; no seal replacement was required at this time. Housing was reassembled and re-torqued to specification, and the pump was returned to service following a post-maintenance vibration check.

Findings

Bearing wear was attributed to extended operation between lubrication cycles. The technician noted in the closing remarks that the standard 90-day lubrication interval specified in SOP-PM-204 may be insufficient for current duty conditions on this pump, and recommended the interval be reviewed. This recommendation was not formally actioned at the time of closure.

Parts and Materials Used

Item	Quantity	Notes
Bearing assembly (P-204 series, OEM part)	1	Full replacement
Housing gasket	1	Replaced during reassembly
Bearing grease (specified grade)	150 g	Applied on reassembly

Post-Maintenance Verification

Measurement Point	Reading (mm/s)	Threshold (mm/s)	Status
Drive-end bearing	2.4	4.5	Normal
Non-drive-end bearing	2.0	4.5	Normal

Sign-Off

Work Completed By	Rajesh Kumar	Date/Time	14-Nov-2024, 16:20
Permit Closed By	Rajesh Kumar	Date/Time	14-Nov-2024, 16:30
Reviewed By	Suresh Patil (Compliance Officer)	Date	15-Nov-2024

This work order is a controlled maintenance record. Filed under Unit 4 Maintenance Records, Process Bay 2 – Pump P-204A.
Note: a subsequent bearing-related failure on this equipment (INC-2025-041, 11-Feb-2025) referenced this work order and the lubrication-interval recommendation noted above during root-cause investigation.